

Unit 3:

RSA Attacks

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1. Prime factorization
2. Cyclic encryption
3. Birthday paradox
4. Side-channel attacks

Summary

- Prime factorization
 - Finding p and q
- Cyclic encryption
 - Find secret without the private key
- Birthday paradox
 - Finding the private key
- Side-channel attacks
 - Take advantage of implementation issues

Prime factorization

Prime factorization

- Goal
 - Obtain the private key d from the public key (e, n)
- Problem to solve
 - Prime factorization
 - There is no efficient algorithm capable of an efficient factorization of large numbers like the ones used in RSA
 - The largest factorized number is RSA-250 with 829 bits (Feb 2020)
- ¿Why would this break RSA?
 - If we can factorize n , we can get p and q
 - Then we can get $\phi(n) = (p-1)(q-1)$.
 - And then we can calculate $d = \text{inv}[e, \phi(n)]$.

Cyclic encryption

Cyclic encryption

- Goal
 - Find the number encrypted with the public key without knowing the private one
- How?
 - We will encrypt the ciphertext with the public key until we get back the original result
 - When we get back the original result we will know that the previous value was the original secret

$$C_i = C_{i-1}^e \bmod n \quad (i = 1, 2, \dots) \quad \text{with } C_0 = C$$

Example

With $p = 13$, $q = 19$, ($n = p \cdot q = \mathbf{247}$). Also, $\phi(n) = (p-1)(q-1) = 216$.
We pick $e = \mathbf{29}$ and obtain a private key $d = \mathbf{149}$, which is **unknown**

A number was encrypting resulting in $C = M^{29} \bmod 247 = 119$

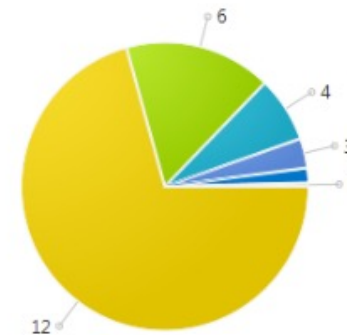
$C_0 = \mathbf{119}$ (captured value)	$C_4 = 175^{29} \bmod 247 = 54$
$C_1 = 119^{29} \bmod 247 = 6$	$C_5 = 54^{29} \bmod 247 = \mathbf{123}$
$C_2 = 6^{29} \bmod 247 = 93$	We don't know this was the secret
$C_3 = 93^{29} \bmod 247 = 175$	$C_6 = 123^{29} \bmod 247 = \mathbf{119}$ (now we do)

We obtained again $C = 119$, so the previous step will be the actual encryption of the original value N .

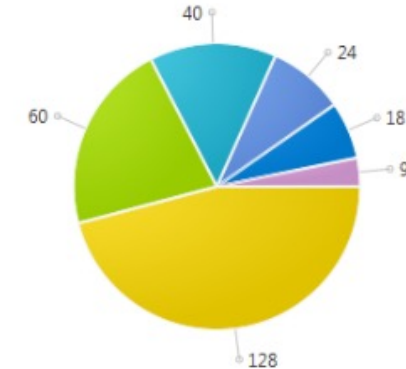
Rings

- When we perform keep encrypting with the public key we form something we call rings
- These rings can have different lengths that go from 0 to $n-1$.
- As an example, with $(n = 2.173, e = 3)$. One ring will be of length 12 : {2, 8, 512, 210, 1.847, 336, 1.168, 538, 1.519, 1.853, 840, 866}

Length	Nº Rings	Prob.
12	128	70,69%
6	60	16,57%
4	40	7,36%
3	24	3,31%
2	18	1,66%
1	9	0,41%



Distribución de probabilidades



Número de anillos

Birthday Paradox

Birthday Paradox

- We only need 23 people to have a 50% chance (or more) of having two of them sharing their birthday
- We say this is a paradox because 23 seems a low number compared to the 365 days of a year
- In general,
 - If the possible values of something are n , we don't need 2^n attempts (or 2^{n-1} on average). We only need to search in $2^{n/2}$, which is much smaller

Attacking RSA with the birthday paradox

1. The attacker picks two numbers i, j ; normally $i = 0$ and $j = [n/2]$.
 - In this way they divide the search in two parts.
3. Then they start calculating $N^i \bmod n$ and $N^j \bmod n$, (with $N=2$ normally) increasing i and j by 1 each time until they find a match.
4. When they find a match they have found two values which encrypt into the same value. These are called paired keys
5. Using the birthday paradox we would need to try $1,2 * n^{1/2}$ values to have a 50% of a match (with n being all possible values of the key in this case)

Example

With $p = 13$ y $q = 19$, $n = p \cdot q = 247$. If $e = 11$, then $d = 59$. The attacker knows $n = 247$ y $e = 11$. For the attack they will use $N = 2$ and pick $i = 0$ and $j = 124$. We will do 17 encryptions in i (from 0 to 16) and 17 encryptions in j (from 124 until 142) to find a match

$i=0$	$2^0 \bmod 247 = 1$
$i=1$	$2^1 \bmod 247 = 2$
$i=2$	$2^2 \bmod 247 = 4$
$i=3$	$2^3 \bmod 247 = 8$
$i=4$	$2^4 \bmod 247 = 16$
$i=5$	$2^5 \bmod 247 = 32$
$i=6$	$2^6 \bmod 247 = 64$
$i=7$	$2^7 \bmod 247 = 128$
$i=8$	$2^8 \bmod 247 = 9$
$i=9$	$2^9 \bmod 247 = 18$
$i=10$	$2^{10} \bmod 247 = 36$
$i=11$	$2^{11} \bmod 247 = 72$
$i=12$	$2^{12} \bmod 247 = 144$
$i=13$	$2^{13} \bmod 247 = 41$

$i=14$	$2^{14} \bmod 247 = 82$
$i=15$	$2^{15} \bmod 247 = 164$
$i=16$	$2^{16} \bmod 247 = 81$
$i=17$	$2^{17} \bmod 247 = 162$
$i=18$	$2^{18} \bmod 247 = 77$

$j=124$	$2^{124} \bmod 247 = 81$
$j=125$	$2^{125} \bmod 247 = 162$
$j=126$	$2^{126} \bmod 247 = 77$
$j=127$	$2^{127} \bmod 247 = 154$
$j=128$	$2^{128} \bmod 247 = 61$
$j=129$	$2^{129} \bmod 247 = 122$
$j=130$	$2^{130} \bmod 247 = 244$
$j=131$	$2^{131} \bmod 247 = 241$
$j=132$	$2^{132} \bmod 247 = 235$
$j=133$	$2^{133} \bmod 247 = 223$
$j=134$	$2^{134} \bmod 247 = 199$
$j=135$	$2^{135} \bmod 247 = 151$
$j=136$	$2^{136} \bmod 247 = 55$
$j=137$	$2^{137} \bmod 247 = 110$

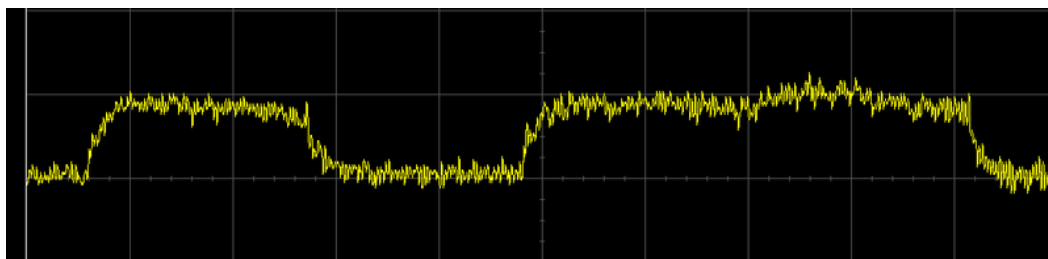
$j=138$	$2^{138} \bmod 247 = 220$
$j=139$	$2^{139} \bmod 247 = 193$
$j=140$	$2^{140} \bmod 247 = 139$
$j=141$	$2^{141} \bmod 247 = 31$
$j=142$	$2^{142} \bmod 247 = 62$

Side-Channel Attacks

Side-Channel Attacks

Kind of attacks

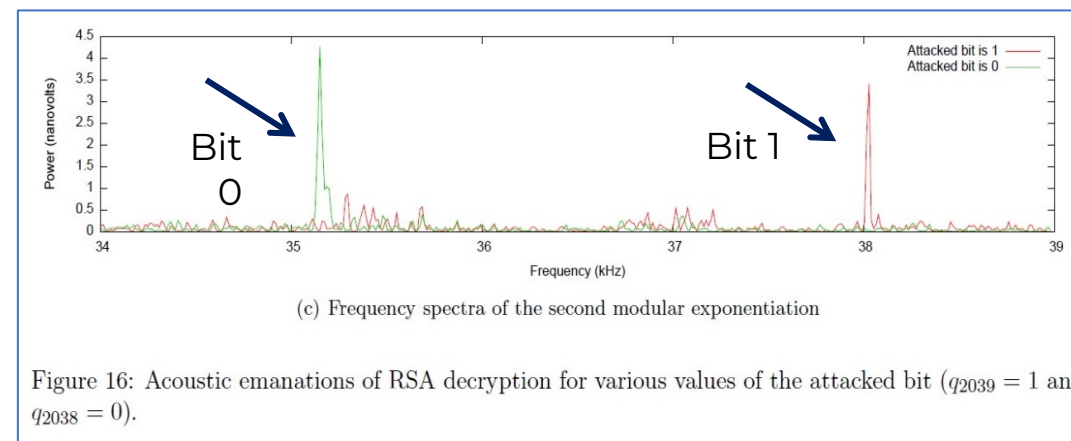
- Energy consumption
- Electromagnetic radiation
- Noise...



Effect of the fast exponentiation algorithm in a PC:
Left is $x^2 \bmod n$, right is $x^2 * A \bmod n$

<https://www.cs.tau.ac.il/~tromer/acoustic/> (Daniel Genkin, Adi Shamir, Eran Tromer)

<https://www.tau.ac.il/~tromer/papers/acoustic-20131218.pdf>



Israel: 2014



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